

USB Drive Wipe Recovery

Do you have a USB drive that has been wiped, maybe by a disgruntled individual? Let's recover the files and information hidden within the formatted USB image in a few easy steps using **Okteta**, a hex editor, and **ExifTool**.

Use a tool like **Win32 Disk Imager** to get the image file (if you want me to show you how to get the image file, DM and I will be happy to show you. I already have the USB Image File)

Open the USB Image File in the hex editor to see what the USB image looks like. We see the disk is not bootable from the disk's first sector, which suggests it has been formatted. What can we get from this????

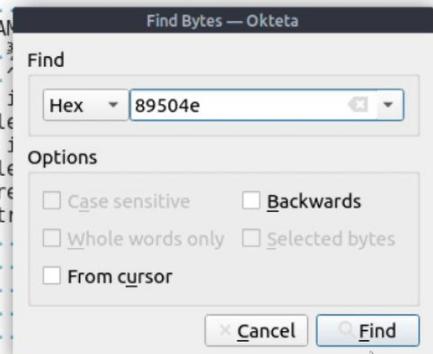
0000000000	EB 58 90 6D	6B 66 73 2E	66 61 74 00	02 08 20 00	ëX.mkfs.fat...
0000000016	02 00 00 00	00 F8 00 00	3F 00 FF 00	00 00 00 00	ø..?.ÿ.....
0000000032	00 00 10 00	00 04 00 00	00 00 00 00	02 00 00 00	
0000000048	01 00 06 00	00 00 00 00	00 00 00 00	00 00 00 00	
0000000064	80 00 29 B6	EB 71 2E 4E	4F 20 4E 41	4D 45 20 20	..)¶ëq.NO NAME
0000000080	20 20 46 41	54 33 32 20	20 20 0E 1F	BE 77 7C AC	FAT32 ..¾w ~
0000000096	22 C0 74 0B	56 B4 0E BB	07 00 CD 10	5E EB F0 32	"Àt.V'.»...Í.^ëö2
0000000112	E4 CD 16 CD	19 EB FE 54	68 69 73 20	69 73 20 0E	äÍ.Í.ëþThis is n
0000000128	6F 74 20 61	20 62 6F 6F	74 61 62 6C	65 20 64 69	ot a bootable di
0000000144	73 6B 2E 20	20 50 6C 65	61 73 65 20	69 6E 73 65	sk. Please inse
0000000160	72 74 20 61	20 62 6F 6F	74 61 62 6C	65 20 66 6C	rt a bootable fl
0000000176	6F 70 70 79	20 61 6E 64	0D 0A 70 72	65 73 73 20	oppy and..press
0000000192	61 6E 79 20	6B 65 79 20	74 6F 20 74	72 79 20 61	any key to try a
0000000208	67 61 69 6E	20 2E 2E 2E	20 0D 0A 00	00 00 00 00	gain
0000000224	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
0000000240	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
0000000256	00 00 00 00	00 00 00 00	00 00 00 00	00 00 00 00	
0000000272	AA AA AA AA	AA AA AA AA	AA AA AA AA	AA AA AA AA	

Use **CTRL + F** to find common file signatures to see if there is any file that can be recovered in the formatted disk image.

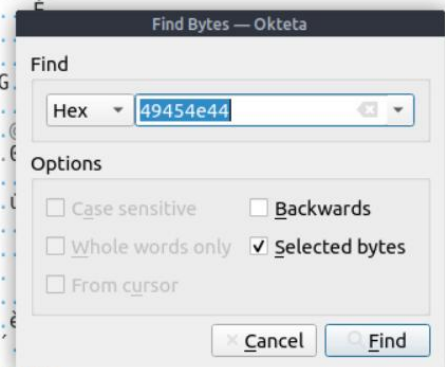
You can look up the **list of file signatures** using <https://filesig.search.org/>
https://en.wikipedia.org/wiki/List_of_file_signatures

So let's search for **PNG file signature**, the PNG file Header is 89 50 4E 47 0D 0A 1A 0A and the Footer is 49 45 4E 44 AE 42 60 82

0000000000 EB 58 90 0D 0B 06 73 2E 0B 01 74 00 02 08 20 00 ex.mKTS.Tat...
0000000016 02 00 00 00 00 F8 00 00 3F 00 FF 00 00 00 00 00ø..?..ÿ....
0000000032 00 00 10 00 00 04 00 00 00 00 00 00 00 00 00
0000000048 01 00 06 00 00 00 00 00 00 00 00 00 00 00 00
0000000064 80 00 29 B6 EB 71 2E 4E 4F 20 4E 41 4D 45 20 20 ..)Tëq.NO NAM
0000000080 20 20 46 41 54 33 32 20 20 20 0E 1F BE 77 7C AC FAT32
0000000096 22 C0 74 0B 56 B4 0E BB 07 00 CD 10 5E EB F0 32 "Ät.V".»..I.
0000000112 E4 CD 16 CD 19 EB FE 54 68 69 73 20 69 73 20 6E äI.Ï.ëpThis
0000000128 6F 74 20 61 20 62 6F 6F 74 61 62 6C 65 20 64 69 ot a bootable
0000000144 73 68 2E 20 20 50 6C 65 61 73 65 20 69 6E 73 65 sk. Please
0000000160 72 74 20 61 20 62 6F 6F 74 61 62 6C 65 20 66 6C rt a bootable
0000000176 6F 70 70 79 20 61 6E 64 0D 0A 70 72 65 73 73 20 oppy and..pre
0000000192 61 6E 79 20 6B 65 79 20 74 6F 20 74 72 79 20 61 any key to tr
0000000208 67 61 69 6E 20 2E 2E 2E 20 0D 0A 00 00 00 00 00 gain ...
0000000224 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000240 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000256 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000272 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000288 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000304 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000320 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000336 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000352 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0000000368 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00



0001526080 40 24 01 11 00 00 00 00 00 00 00 80 4C 48 04 7F rpa...LNU
0001526096 01 00 00 00 00 00 00 B2 41 08 68 00 00 00 00 00 ..²A.h
0001526112 00 80 8C 48 22 55 02 06 00 00 00 00 00 00 C8 02 'H"U
0001526128 09 60 00 00 00 00 00 00 80 8C 90 00 06 00 00 00 É.
0001526144 00 00 00 C8 08 09 60 00 00 00 00 00 80 8C C8 É.
0001526160 0F 7A 00 00 00 00 00 00 00 00 F4 47 92 6A 01 0D .z...ôG
0001526176 00 00 00 00 00 00 90 09 12 C0 00 00 00 00 00 00 ...Ä
0001526192 00 19 91 84 04 30 00 00 00 00 00 00 40 26 24 CA ...ö
0001526208 BF 00 00 00 00 00 00 00 D9 90 84 0A 30 00 00 00 ÿ...Ü
0001526224 00 00 00 40 26 24 A9 0A 30 00 00 00 00 00 00 40 @ç\$@.ö
0001526240 26 E4 07 3D 00 00 00 00 00 00 00 00 FA 23 49 53 ää.=...
0001526256 09 60 00 00 00 00 00 00 80 2C 90 00 06 00 00 00 É.Ä.Ä
0001526272 00 00 00 C8 88 C4 12 C0 00 00 00 00 00 00 00 D9 ...Ä
0001526288 90 A4 2A C0 00 00 00 00 00 00 00 99 20 01 0C 00 ...Ä
0001526304 00 00 00 00 00 90 11 12 C0 00 00 00 00 00 00 00 ...ö
0001526320 19 91 1F F4 00 00 00 00 00 00 00 00 E8 8F 24 15 ...É
0001526336 00 06 00 00 00 00 00 00 00 C8 84 24 B4 80 06 00 00 ...É.\$
0001526352 00 00 00 00 00 00 00 00 00 C8 04 09 60 00 00 80 8C ...É.
0001526368 90 00 06 00 00 00 00 00 00 C8 88 44 FD 17 00 00 ...É.Dý
0001526384 00 00 00 00 20 1B 12 F5 5F 00 00 00 00 00 00 80 ...ö
0001526400 6C D0 02 1A 00 00 00 00 00 00 20 23 FE 3F F7 1D lD...#b?÷
0001526416 62 41 50 89 3A 16 00 00 00 00 49 45 4E 44 AE 42 bAP...IEND@B
0001526432 60 82 00 00 00 00 00 00 00 00 00 00 00 00 00



0001069056 89 50 4E 47 0D 0A 1A 0A 00 00 00 0D 49 48 44 52 .PN.....IHDR
0001069072 00 00 07 80 00 00 04 38 08 06 00 00 00 E8 D3 C18.....èÖÄ
0001069088 43 00 00 00 09 70 48 59 73 00 00 0B 12 00 00 0B C....pHYs.....
0001069104 12 01 D2 DD 7E FC 00 00 00 9C 7A 54 58 74 52 61 ..Öÿ~ü....zTXtRa
0001069120 77 20 70 72 6F 66 69 6C 65 20 74 79 70 65 20 69 w profile type i
0001069136 70 74 63 00 00 78 9C 3D 4E 39 0A 03 31 0C EC FD ptc..x.=N9..1.ÿ
0001069152 8A 7D 82 2D 8D 46 76 9D 2A 5D 8A 7D C1 FA 80 40 .}.-.Fv.*].}Áú.@
0001069168 20 21 FF 2F A2 DD 22 23 04 62 0E 34 E9 FE D8 6F !ÿ/çÿ"#.b.4épðo
0001069184 DB E7 FB 5E CF D7 4C DB 05 AF 49 2B 04 0D 23 23 Üçû^Ï×LÜ.~I+..##
0001069200 E6 0F 94 D2 B3 78 30 6A 52 A9 D2 54 34 C7 42 B2 æ..Ö³x0jR©ÖT4ÇB²


```

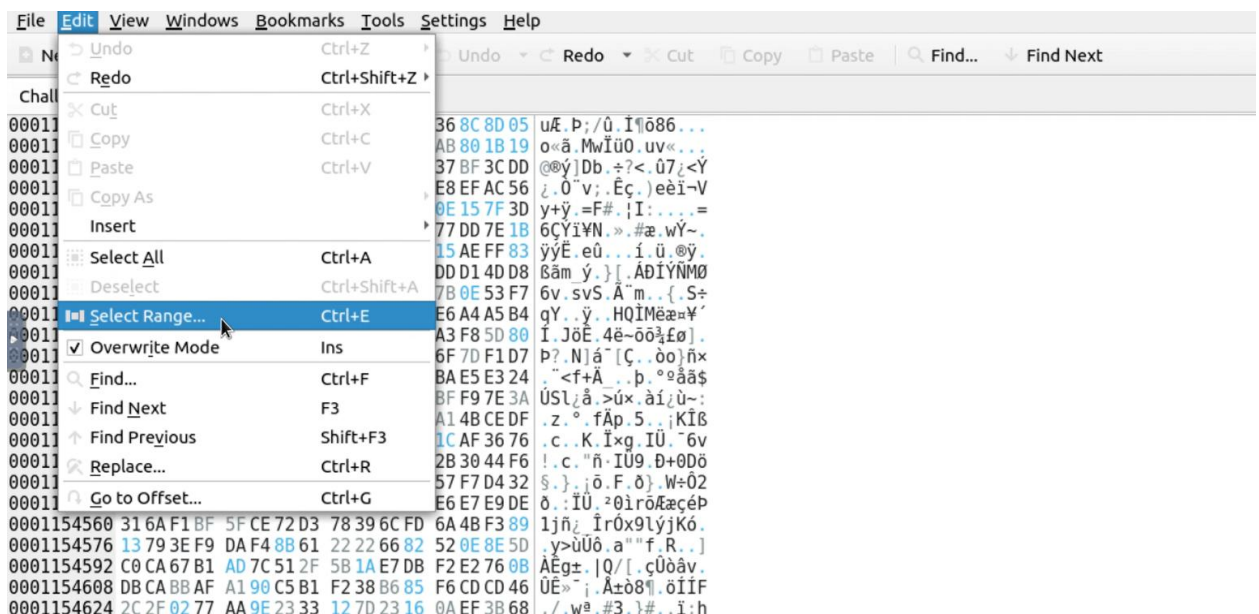
0001526064 00 00 00 00 00 b4 81 04 30 00 00 00 00 00 00 40 .....d..@
0001526080 46 24 61 11 60 00 00 00 00 00 00 80 4C 48 D4 7F F$a.`...LH0.
0001526096 01 00 00 00 00 00 00 B2 41 0B 68 00 00 00 00 00 .....^A.h.
0001526112 00 80 8C 48 22 55 02 06 00 00 00 00 00 00 C8 02 ..H"U.....E.
0001526128 09 60 00 00 00 00 00 00 80 8C 90 00 06 00 00 00 `..E.....E
0001526144 00 00 00 C8 08 09 60 00 00 00 00 00 80 8C C8 ..E.`.....E
0001526160 0F 7A 00 00 00 00 00 00 00 00 F4 47 92 6A 01 0D .z.....ôG.j..
0001526176 00 00 00 00 00 00 90 09 12 C0 00 00 00 00 00 00 .....Ä.....
0001526192 00 19 91 84 04 30 00 00 00 00 00 00 40 26 24 CA .....0.....@&$E
0001526208 BF 00 00 00 00 00 00 00 D9 90 84 0A 30 00 00 00 .;.U.....0
0001526224 00 00 00 40 26 24 A9 0A 30 00 00 00 00 00 00 40 .....@&$0.0.....@
0001526240 26 E4 07 3D 00 00 00 00 00 00 00 00 FA 23 49 53 &ä.=.....ú#IS
0001526256 09 60 00 00 00 00 00 00 80 2C 90 00 06 00 00 00 `..E.Ä.Ä.....Ü
0001526272 00 00 00 C8 88 C4 12 C0 00 00 00 00 00 00 D9 ..E.Ä.Ä.....Ü
0001526288 90 A4 2A C0 00 00 00 00 00 00 00 99 20 01 0C 00 .π*Ä.....
0001526304 00 00 00 00 00 90 11 12 C0 00 00 00 00 00 00 00 .....Ä.....
0001526320 19 91 1F F4 00 00 00 00 00 00 00 00 E8 8F 24 15 ...ô.....è$.
0001526336 00 06 00 00 00 00 00 00 C8 84 24 B4 80 06 00 00 ..E.$'.....
0001526352 00 00 00 00 C8 04 09 60 00 00 00 00 00 80 8C ...E.....E.Dý..
0001526368 90 00 06 00 00 00 00 00 00 C8 88 44 FD 17 00 00 .....E.Dý..
0001526384 00 00 00 00 20 1B 12 F5 5F 00 00 00 00 00 00 80 .....ô.....
0001526400 6C D0 02 1A 00 00 00 00 00 00 20 23 FE 3F F7 1D lD.....#p?÷.
0001526416 62 41 50 89 3A 16 00 00 00 00 49 45 4E 44 AF 42 bAP:.....IEND@B
0001526432 60 82 00 00 00 00 00 00 00 00 00 00 00 00 00 .....

```

Searching the PNG header with 89504E47, the header starts at the offset of 0001069056 and searching the PNG footer with 49454E44, the footer starts at the offset of 0001526432, there is an indication of a PNG image.

To extract the data,

Option 1: Select the range of the data, copy the range of the data using the offset where the data was found as start and end offset. Convert the offset into Decimal to get accurate data as expected. Save the data range using .png extension.



Option 2: Extract the file using the dd command, in the dd command, specify the disk location, where the recovered image is to be saved, the offset value alongside the difference between the start and end offset.

```
File Edit View Search Terminal Help
~/Desktop/Carving_Challenges$ dd if=Challenge1_Manual_Carve_usb
.img of=Image.png bs=1 skip=134483968 count=463
463+0 records in
463+0 records out
463 bytes copied, 0.00307787 s, 150 kB/s
~/Desktop/Carving_Challenges$ ls
Carving_Challenges.dd  Challenge2_clock_spinning  Image.png
Challenge1_Manual_Carve_usb.img  Challenge2_deleted_disk.img
~/Desktop/Carving_Challenges$
```

The file will be extracted and saved based on the name provided as above.

ExifTool can be used to extract the metadata of the image.

Remember, for forensic investigations, it is crucial to **perform any experiments on a sandbox or isolated host** so as not to infect live hosts with unsuspecting malware that may be hidden by an adversary on recovered files.